A TREE AUTOMATON WITH INFINITELY MANY ROGUE FIXED POINTS UNDER EVERY SUPERCRITICAL OFFSPRING LAW

YITZCHAK SHMALO

Abstract. Johnson, Podder and Skerman asked whether a tree automaton with finitely many, but at least three, states can have infinitely many fixed points of its automaton distributional map. Under binary branching an example with three states is implicit in earlier work of Rozikov and Zada on quadratic stochastic operators. We show that the phenomenon holds for every supercritical offspring law, Poisson laws included. For every $k \geq 2$ there is an automaton on $k+1$ states whose fixed points, for an offspring law with generating function G and mean greater than one, are one point mass together with the $(k-1)$-dimensional simplex of laws that give the distinguished state mass s_*, where $G'(s_*) = 1$. Every point of the simplex is rogue, with no moment assumption on the offspring law; for the points other than the k vertices this has a short self-contained proof.

1. Introduction

Let Σ be a finite set of states. Following [4], a *tree automaton* is a map $A\colon \mathbb{N}_0^{\Sigma} \to \Sigma$: the state of a parent is $A(\vec{n})$, where $\vec{n} = (n_\sigma)_{\sigma\in\Sigma}$ counts its children in each state. Let χ be a law on $\mathbb{N}_0$ with $\chi(\{2,3,\dots\}) > 0$, the standing assumption of [4, Section 1.2], and let Δ be the simplex of laws on Σ. The *automaton distributional map* $\Psi\colon \Delta \to \Delta$ sends x to the law of $A(\vec{n})$, where the root has $N \sim \chi$ children whose states are independent with law x [4, Section 1.5]. An *interpretation* of A under $\mathrm{GW}(\chi)$ is a measurable map ι from locally finite, ordered, rooted trees (with their Borel σ-field) to Σ such that, for almost every $\mathrm{GW}(\chi)$ tree t, the colouring $v \mapsto \iota(t(v))$ is compatible with A; the law of $\iota(T)$ is then a fixed point of Ψ, and a fixed point that does not arise this way is called *rogue* [4, Sections 1.4 and 1.5].

Johnson, Podder and Skerman remark that the case $|\Sigma| = 2$ can be ruled out by arguing that the automaton distributional map is analytic. The argument needs a hypothesis on χ: Remark 2.1 gives a two-state automaton whose map is the identity. Fixed points of such maps are solutions of recursive distributional equations in the sense of the survey of Aldous and Bandyopadhyay [1], and two-state examples arise as the outcome probabilities of the games on Galton–Watson trees studied by Holroyd and Martin [3]. Johnson, Podder and Skerman asked [4, Section 6]: when $3 \leq |\Sigma| < \infty$, can an automaton have infinitely many fixed points?

For binary branching the answer was in effect already known. Rozikov and Zada [5, Theorem 2(iv) and equation (21)] found a segment of fixed points for a family of quadratic stochastic operators on the two-dimensional simplex; Ganikhodzhaev, Mukhamedov and Rozikov [2] survey these operators. At their parameters $a = b = 0$, $c = 1$ the operator is $(x, y, z) \mapsto (2xz,\ 2yz,\ z^2+(x+y)^2)$; its heredity coefficients are 0 or 1, so it is the automaton distributional map of a deterministic three-state automaton under $\mathrm{GW}(\delta_2)$, namely the automaton A_2 below,

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and its fixed segment is $x + y = 1/2$. What follows extends this example to every supercritical offspring law ([4] treats general offspring laws, and most of its examples are Poisson) and proves that every point of the simplex is rogue.

Fix $k \geq 2$ and let $\Sigma_k = \{a, b_1, \dots, b_k\}$. Call a the *even* state and the b_i the *odd* states. Let A_k be the automaton whose output is b_i when the vertex has exactly one odd child and that child is in state b_i, and a otherwise:

$$A_k(\vec{n}) = \begin{cases} b_i & \text{if } n_{b_i} = 1 \text{ and } n_{b_j} = 0 \text{ for } j \neq i, \\ a & \text{otherwise.} \end{cases}$$

The number n_a of even children plays no role, and a leaf is in state a. On a vertex with two children, A_k is the symmetric rule

$$g(a,a) = a, \qquad g(b_i, b_j) = a, \qquad g(a, b_i) = g(b_i, a) = b_i,$$

and on a vertex with one child it copies that child.

Write $G(s) = \sum_{n \geq 0} \chi(n) s^n$ for the generating function of χ and $m = G'(1) \in (0, \infty]$ for its mean. Under the standing assumption $\chi(\{2, 3, \dots\}) > 0$, G' is continuous and strictly increasing on $[0, 1)$, with $G'(0) = \chi(1) < 1$. So when $m > 1$ there is a unique $s_* \in (0, 1)$ with $G'(s_*) = 1$.

Theorem 1.1. *If $m > 1$, the fixed points of the automaton distributional map of A_k under $\mathrm{GW}(\chi)$ are exactly the point mass δ_a and the laws x with $x(a) = s_*$. The latter form a $(k-1)$-dimensional simplex, whose relative interior consists of laws of full support. If $m \leq 1$, the only fixed point is δ_a.*

Corollary 1.2. *For $\chi = r\delta_1 + (1-r)\delta_2$ with $0 \leq r < 1$, $s_* = 1/2$; for $k = 2$ the three-state automaton A_2 then has the line of fixed points $(\frac{1}{2}, \tau, \frac{1}{2} - \tau)$, $0 \leq \tau \leq \frac{1}{2}$, in the order (a, b_1, b_2). For the Poisson law with mean $\lambda > 1$, $s_* = 1 - (\log \lambda)/\lambda$.*

Proof. In the first case $G'(s) = r + 2(1-r)s$, and in the second $G'(s) = \lambda e^{\lambda(s-1)}$; solve $G'(s) = 1$. □

Theorem 1.3. *Let $m > 1$ and let ι be an interpretation of A_k under $\mathrm{GW}(\chi)$. Then the law of $\iota(T)$ is δ_a or $s_*\delta_a + (1 - s_*)\delta_{b_i}$ for some i. Hence every fixed point other than these $k + 1$ laws is rogue; in particular so is every fixed point of full support.*

Theorem 1.4. *If $m > 1$, then δ_a is the only fixed point of A_k under $\mathrm{GW}(\chi)$ that admits an interpretation.*

Proposition 1.5. *Under $\mathrm{GW}(\delta_2)$ the only interpretation of A_k, up to almost sure equality, is the constant map with value a. Hence every fixed point with $x(a) = 1/2$ is rogue.*

Thus for every supercritical offspring law the set of fixed points contains a continuum of rogue laws. Theorem 1.4 contains Theorem 1.3 and Proposition 1.5; we keep them because their proofs are elementary, while the proof of Theorem 1.4 goes through the spine argument of [4, Section 4].

2. Proofs

Proof of Theorem 1.1. Let x be a law on Σ_k and write $s = x(a)$. Given $N = n$, the children are independent with law x, and the root is in state b_i exactly when one child is b_i and the other $n - 1$ are a; this has probability $n\, x(b_i) s^{n-1}$. Summing over n,

$$\Psi(x)(b_i) = x(b_i)\, G'(s), \qquad \Psi(x)(a) = 1 - (1 - s)\, G'(s),$$

for $s < 1$, where $G'(s)$ is finite. (A root with no children is in state a, and the term $n = 0$ contributes nothing to the first formula.)

If $x(b_i) = 0$ for every i, then $x = \delta_a$ and $\Psi(x) = \delta_a$. Otherwise $s < 1$, and a fixed point must satisfy $G'(s) = 1$, from the coordinate of any b_i with $x(b_i) > 0$. Conversely, if $G'(s) = 1$ then $\Psi(x)(b_i) = x(b_i)$ for every i and $\Psi(x)(a) = 1 - (1 - s) = s$, so x is fixed. For $s < 1$ we have $G'(s) < G'(1) = m$, so $G'(s) = 1$ has no solution when $m \le 1$ and exactly the solution s_* when $m > 1$. The laws with $x(a) = s_*$ are $s_*\delta_a + (1 - s_*)\mu$ with μ any law on $\{b_1, \dots, b_k\}$, a $(k-1)$-simplex, and x has full support exactly when μ does. □

For $\chi = \delta_2$ and $k = 2$, writing (x_0, x_1, x_2) for the masses of (a, b_1, b_2), the map is

$$\Psi(x_0, x_1, x_2) = \big(x_0^2 + (x_1 + x_2)^2,\ 2x_0x_1,\ 2x_0x_2\big),$$

and the line in Corollary 1.2 can be checked by substitution.

Proof of Proposition 1.5. Under $\mathrm{GW}(\delta_2)$ the tree is almost surely the infinite binary tree t_*, and every subtree $t_*(v)$ equals t_*. An interpretation ι therefore colours every vertex of t_* with the single state $\sigma = \iota(t_*)$, and compatibility at a vertex with two children in state σ requires $\sigma = g(\sigma, \sigma)$. This holds for $\sigma = a$ and fails for $\sigma = b_i$, since $g(b_i, b_i) = a$. Conversely the constant map a is compatible at every vertex. So the law of $\iota(T)$ is δ_a, and every fixed point with $x(a) = 1/2$ is rogue. □

Proof of Theorem 1.3. Let x be the law of $\iota(T)$; it is a fixed point [4, Section 1.5], so by Theorem 1.1 either $x = \delta_a$ or $s := x(a) = s_*$. Assume the latter. Call a tree t *odd* if $\iota(t) \neq a$ and *even* otherwise, and let O be the set of odd trees, so $\mathbb{P}(T \in O) = 1 - s_* > 0$. Almost surely the colouring $v \mapsto \iota(T(v))$ is compatible with A_k at every vertex. By the definition of A_k, an odd vertex v then has exactly one odd child $\hat{v}$, and $\iota(T(\hat{v})) = \iota(T(v))$.

For an odd tree t with this property at its root, let $\theta t = t(\hat{o})$ be the subtree of the odd child $\hat{o}$ of the root o and let $\xi(t)$ record the rest of the first generation: the number n of children of the root, the position j of the odd child among them, and the $n-1$ even subtrees of its siblings, in order. Given $N = n$ the subtrees of the children of the root are independent $\mathrm{GW}(\chi)$ trees, and whether each is odd is decided by that subtree alone. Hence, for Borel sets B of trees and C of decorations,

$$\mathbb{P}(T \in O,\ \theta T \in B,\ \xi(T) \in C) = \mathbb{P}(T \in B \cap O)\,\nu(C),$$

where $\nu(C) = \sum_n \chi(n) \sum_{j=1}^{n} \mathbb{P}((n, j, T'_1, \dots, T'_{n-1}) \in C,\ T'_1, \dots, T'_{n-1} \notin O)$ with $T'_1, T'_2, \dots$ independent $\mathrm{GW}(\chi)$ trees. The total mass of ν is $\sum_n \chi(n)\, n\, s_*^{n-1} = G'(s_*) = 1$. Dividing by $\mathbb{P}(T \in O)$: under $\mathbb{P}(\,\cdot \mid T \in O)$, θT is odd with the conditional law of T, and $\xi(T)$ has law ν and is independent of θT.

Set $T_0 = T$, $T_{r+1} = \theta T_r$ and $\xi_r = \xi(T_r)$; these are defined almost surely on $\{T \in O\}$, since every vertex of the odd ray $o = v_0, v_1, v_2, \dots$, where $v_{r+1} = \hat{v}_r$, is odd. Iterating the previous paragraph, under $\mathbb{P}(\,\cdot \mid T \in O)$ the ξ_r are independent with law ν, and $\xi_0, \dots, \xi_{r-1}$ are independent of T_r. The tree T_r is the odd ray from its root with the decorations $\xi_r, \xi_{r+1}, \dots$ hung along it; its first h generations are determined by $\xi_r, \dots, \xi_{r+h-1}$, so T_r is a measurable function of $(\xi_r, \xi_{r+1}, \dots)$. Fix i. For each r let $E_r \in \sigma(\xi_r, \xi_{r+1}, \dots)$ be the event that the measurable function of $(\xi_r, \xi_{r+1}, \dots)$ giving T_r takes a value t with $\iota(t) = b_i$. Since ι is constant along the odd ray, each E_r agrees with $\{\iota(T) = b_i\}$ up to a null set of $\mathbb{P}(\,\cdot \mid T \in O)$. Hence $E = \liminf_{r\to\infty} E_r$, which lies in $\sigma(\xi_j : j \ge h)$ for every h, agrees with $\{\iota(T) = b_i\}$ up to a null set, and by Kolmogorov's zero-one law $\mathbb{P}(\iota(T) = b_i \mid T \in O) \in \{0, 1\}$. Thus x restricted to $\{b_1, \dots, b_k\}$ is concentrated on a single state. □

Proof of Theorem 1.4. Merge $b_1, \dots, b_k$ into one state b. The result is the two-state automaton B that outputs b when exactly one child is in state b, and a otherwise. If ι interprets a fixed point x of A_k with $x \neq \delta_a$, then composing ι with the merging map interprets the fixed point $\bar{x} = (s_*, 1 - s_*)$ of B. So it suffices to show that $\bar{x}$ is rogue. By [4, Theorem 1.7] this holds if the pivot tree of $\bar{x}$ is supercritical, provided χ has a finite logarithmic moment; we first check that for B the moment is not needed, and then compute the pivot tree.

The direction of [4, Theorem 1.7] used here is proved in [4, Section 4]. The logarithmic moment is used substantively only in [4, Lemma 4.14], through [4, Lemma 4.13], and [4, Lemma 4.15] inherits it from there. In [4, Lemma 4.14] it serves to bound the number of children of the root of T^{0s} and of T^{1s}, where T^{ℓ} is the random state tree conditioned on the label ℓ of its root, and $\ell = 0s$ or $1s$ means that the root has the colour 0 or 1 and starts an infinite path of pivotal vertices [4, Section 4]. For B, with the colours a and b in the roles of 0 and 1, write $s = s_*$ and $p = 1 - s_*$, and let N be the number of children of the root. Then $\mathbb{P}(N = n,\ \text{root in state } b) = \chi(n)\, n\, p\, s^{n-1}$, since the root then has exactly one child in state b. A root labelled $0s$ is in state a and has a pivotal child, so, as shown below, it has either zero or exactly two children in state b, and $\mathbb{P}(N = n,\ \text{label } 0s) \le \chi(n)\big(s^n + \binom{n}{2} p^2 s^{n-2}\big)$. The labels $0s$ and $1s$ have positive probability when the pivot tree is supercritical [4, Proposition 2.6(c)], so dividing by $\mathbb{P}(0s)$ and $\mathbb{P}(1s)$,

$$\mathbb{P}(N = n \mid 0s) \le \frac{\chi(n)\big(s^n + \binom{n}{2} p^2 s^{n-2}\big)}{\mathbb{P}(0s)}, \qquad \mathbb{P}(N = n \mid 1s) \le \frac{\chi(n)\, n\, p\, s^{n-1}}{\mathbb{P}(1s)},$$

For $n \ge 1$ both right sides are at most $C\, n^2 s^{n-2}$ for a constant C, since $\chi(n) \le 1$, and both conditional probabilities vanish at $n = 0$, because a root that starts an infinite path of pivotal vertices has a child. As $s < 1$, the root degrees D_{0s} and D_{1s} of T^{0s} and T^{1s} satisfy $\mathbb{E}\, e^{\alpha D_{0s}} < \infty$ and $\mathbb{E}\, e^{\alpha D_{1s}} < \infty$ whenever $e^{\alpha} s < 1$, and in particular have all moments; put $L = \max\{\mathbb{E}D_{0s}, \mathbb{E}D_{1s}\} < \infty$.

With these bounds the conclusion of [4, Lemma 4.14] holds without the moment assumption. In the proof of [4, Lemma 4.14], [4, Proposition 4.9(c)] is applied to the spine vertex V_i conditional on the σ-field $\mathcal{G}$ of the whole spine. That proposition concerns the vertices $V_{i,j}$ off the spine, and the whole spine carries information about the descendants of V_i; the law needed there is the one given the finite spine [4, Proposition 4.9(d)]. We therefore use a finite exploration of the spine, in the notation of [4, Section 4]. Work with the unswitched tree under RST_{0s} with its four labels $0d, 0s, 1d, 1s$, and choose each next spine vertex uniformly among the pivotal children labelled $0s$ or $1s$, using uniform random variables $U_0, U_1, \dots$ independent of each other and of the tree. Let $\mathcal{H}_0$ be generated by the label of the root, and let $\mathcal{H}_{i+1}$ be generated by $\mathcal{H}_i$, the ordered list of the children of V_i with their labels, U_i and the selected child V_{i+1}. Nothing below an unselected child, and nothing below V_{i+1} except its label, is revealed. Conditional on $\mathcal{H}_i$, the unswitched labelled subtree at V_i has law RST_{ℓ_i}, where $\ell_i \in \{0s, 1s\}$ is the label of V_i. This holds for $i = 0$, and it passes from i to $i + 1$ because the subtrees of the children of V_i are independent given their labels, by the multitype branching property [4, Proposition 4.1(c)], and the choice of V_{i+1} uses only these labels and U_i. By the identity $W_i = 1/N(V_i, T^{0s}, \omega_*^{0s})$ at the start of the proof of [4, Lemma 4.12], W_i is $\mathcal{H}_{i+1}$-measurable, so $\Pi_i = W_0 \cdots W_{i-1}$ is $\mathcal{H}_i$-measurable. The conditional law of the subtree at V_i gives $\mathbb{E}[\deg V_i \mid \mathcal{H}_i] \le L$, and the two-step argument of [4, Lemma 4.12], run with this conditional law, gives $\mathbb{E}[W_i W_{i+1} \mid \mathcal{H}_i] \le c$ for a constant $c < 1$, hence $\mathbb{E}\Pi_i \le c^{\lfloor i/2 \rfloor}$. Since $N(V_i, T^{0s}, \omega_*^{0s \to 1s}) \le \deg V_i$,

$$\mathbb{E} \sum_{i \ge 0} \Pi_i\, N(V_i, T^{0s}, \omega_*^{0s \to 1s}) \le \sum_{i \ge 0} \mathbb{E}\big[\Pi_i\, \mathbb{E}[\deg V_i \mid \mathcal{H}_i]\big] \le L \sum_{i \ge 0} c^{\lfloor i/2 \rfloor} < \infty,$$

so the series of [4, Lemma 4.14] is finite almost surely. The subtrees off the spine are still treated with the whole-spine σ-field $\mathcal{G}$, as in [4, Proposition 4.9(b),(c)] and [4, Lemma 4.11]; only the estimate along the spine has changed. The finite series is exactly what [4, Lemma 4.15] takes from [4, Lemma 4.14], so [4, Lemma 4.15] and then [4, Lemmas 4.5 and 4.3] show that the law of the switched tree is absolutely continuous with respect to RST_{1s}. As at the end of [4, Section 4], this contradicts the existence of an interpretation, since switching the colours along the spine leaves the underlying tree unchanged and changes the colour of the root. No moment assumption on χ enters these steps.

In the random state tree of $\bar{x}$ [4, Section 1.6], a vertex is pivotal when changing its colour, and recolouring its ancestors by B, changes the colour of the root; the pivot tree is a two-type Galton–Watson tree, the type being the colour [4, Proposition 2.4]. Given the colour of a vertex, its children are N independent colours with law $\bar{x}$, conditioned on B producing that colour. A vertex in state b has exactly one child in state b, and changing any one child's colour leaves it with zero or two such children, so every child is pivotal. A vertex in state a with no child in state b has all its children pivotal, all in state a; one with exactly two children in state b has those two pivotal and no others; one with three or more has none. Taking expectations, with $\mathbb{P}(b) = pG'(s) = p$ and $\mathbb{P}(a) = s$, the mean offspring matrix of the pivot tree, rows and columns ordered (b, a), is

$$M = \begin{pmatrix} 1 & s\,G''(s) \\ p^2 G''(s)/s & 1 \end{pmatrix} :$$

from b, one pivotal child of type b and $\sum_n \chi(n)\, n(n-1)s^{n-1} = sG''(s)$ of type a on average; from a, $\sum_n \chi(n)\, n\, s^n/s = G'(s) = 1$ of type a and $2\sum_n \chi(n)\binom{n}{2}p^2 s^{n-2}/s = p^2 G''(s)/s$ of type b. Since $\chi(\{2, 3, \dots\}) > 0$ we have $G''(s) \in (0, \infty)$, and the spectral radius of M is $1 + pG''(s) > 1$. □

Remark 2.1. Two states can suffice when χ has infinite mean. Let $\chi(n) = 1/(n(n-1))$ for $n \geq 2$, which has infinite mean and a finite logarithmic moment, and let the two-state automaton on $\{0, 1\}$ output 1 exactly when at most one child is in state 0. If each child is in state 1 with probability q, then

$$\Psi(q) = \sum_{n \geq 2} \frac{q^n + n(1-q)q^{n-1}}{n(n-1)} = \sum_{n \geq 2} \Big(\frac{q^{n-1}}{n-1} - \frac{q^n}{n} \Big) = q,$$

so every law on $\{0, 1\}$ is fixed. If instead $\chi(0) > 0$ and G is entire, as for Poisson laws, three states are needed: Ψ is then analytic on a neighbourhood of $[0, 1]$, and it is not the identity, since a leaf in state 0 gives $\Psi(1) \leq 1 - \chi(0)$ and a leaf in state 1 gives $\Psi(0) \geq \chi(0)$; so Ψ has finitely many fixed points in $[0, 1]$.

The argument in the proof of Theorem 1.4 applies to any two-state automaton and nondegenerate fixed point: under the other hypotheses of [4, Theorem 1.7], including full support of the fixed point, a supercritical pivot tree makes the fixed point rogue whenever the root of T^{0s} and of T^{1s} has a number of children with a finite logarithmic moment, whatever χ is. Indeed, with Π_i as in that proof, $\mathbb{E}\Pi_i \leq c^{\lfloor i/2 \rfloor}$, Markov's inequality and the Borel–Cantelli lemma give $\Pi_i \leq \beta^i$ for all large i, almost surely, for any $\beta \in (\sqrt{c}, 1)$. Choose $\varepsilon > 0$ with $\beta e^{\varepsilon} < 1$. By the conditional law of the subtree at V_i given $\mathcal{H}_i$, $\mathbb{P}(\deg V_i > e^{\varepsilon i}) \leq \mathbb{P}(D_{0s} > e^{\varepsilon i}) + \mathbb{P}(D_{1s} > e^{\varepsilon i})$, and these bounds are summable in i when $\mathbb{E}\log(1 + D_{0s})$ and $\mathbb{E}\log(1 + D_{1s})$ are finite. So $\deg V_i \leq e^{\varepsilon i}$ for all large i, and $\sum_i \Pi_i \deg V_i < \infty$ almost surely; no independence of the degrees along the spine is needed. For Poisson offspring with mean $\lambda > 1$ the spectral radius in the proof of Theorem 1.4 is $1 + \log \lambda$.

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Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Code and data availability

The code and data for this paper are at `https://github.com/yspennstate/automaton-fixed-simplex`; the program `code/check_automaton.py` there checks Theorem 1.1, the matrix M and the identity in Remark 2.1 in exact arithmetic for several offspring laws. The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [7].

Declaration of generative AI and AI-assisted technologies

An OpenAI GPT model proposed the first example, a four-state automaton with a line of fixed points, and pointed to the example of Rozikov and Zada, the reduction used for Theorem 1.4, the two-state example in Remark 2.1 and the form of the series estimate. Claude Opus 5.5 (Anthropic) found the automaton presented here, extended it to general offspring laws, removed the logarithmic moment assumption from Theorem 1.4 and wrote the paper. As a referee, an OpenAI GPT model supplied the finite exploration of the spine and the form of the tail-event argument. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Appendix A. Notation

Symbols are listed in the order in which they first appear. Notation of [4] used only inside one step of a proof is not listed.

Symbol	Meaning	First appears
Σ, Δ	a finite set of states and the simplex of laws on it	Section 1
A, $\vec{n}$	a tree automaton and the numbers of children in each state	Section 1
χ, N	the offspring law and the number of children of the root	Section 1
$\mathrm{GW}(\chi)$, T, $t(v)$	the Galton–Watson law, a tree with that law, the subtree of t rooted at v	Section 1
Ψ, x	the automaton distributional map and a law on the states	Section 1
ι	an interpretation	Section 1
Σ_k, a, b_i	the states of A_k: the even state and the k odd states	Section 1
A_k, g	the automaton of this paper and its rule at a vertex with two children	Section 1
G, m	the generating function and the mean of χ	Section 1
s_*	the solution in $(0,1)$ of $G'(s)=1$	Section 1
τ, λ	the parameter of the line of fixed points and the mean of a Poisson law	Corollary 1.2
s, μ	the mass $x(a)$ and the law on $\{b_1,\dots,b_k\}$ with $x = s_*\delta_a + (1-s_*)\mu$	Section 2
t_*, σ	the infinite binary tree and the state an interpretation gives it	Proposition 1.5
O, $\hat{v}$	the set of odd trees and the odd child of an odd vertex v	Theorem 1.3
θ, ξ, ν	the subtree of the odd child, the rest of the first generation, and its law	Theorem 1.3
v_r, T_r, ξ_r	the odd ray, the subtree at v_r, and its first generation off the ray	Theorem 1.3
h, E_r, E	a number of generations and the events of the zero-one argument	Theorem 1.3
B, b, $\bar{x}$	the automaton with the odd states merged, the merged state, the merged fixed point	Theorem 1.4
p	$1-s_*$	Theorem 1.4
$0s$, $1s$, T^{0s}, T^{1s}	the labels of a vertex of colour a or b that starts an infinite pivotal path, and the random state tree with root so labelled [4, Section 4]	Theorem 1.4
C, D_{0s}, D_{1s}, L	a constant, the root degrees of T^{0s} and T^{1s}, and the larger of their means	Theorem 1.4
V_i, W_i, $\mathcal{G}$	the spine, its weights, and the σ-field of the whole spine [4, Section 4]	Theorem 1.4
$\mathcal{H}_i$, U_i, Π_i	the finite exploration of the spine, the variables that choose the spine, and $W_0\cdots W_{i-1}$	Theorem 1.4
c	the contraction constant of [4, Lemma 4.12]	Theorem 1.4
M	the mean offspring matrix of the pivot tree of $\bar{x}$	Theorem 1.4
q, β, ε	the probability of state 1, and the constants of the logarithmic-moment argument	Remark 2.1

Entries named by a theorem or proposition first appear in its proof.

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Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel
Email address: yitzchak.shmalo@gmail.com